

Table II: RFID Modules from some Manufacturers

Specifications	Texas Instruments		DLP Design Inc		FEIG Electronic		Pepperl + Fuchs	
	RI-RFM-003C	RI-SMD-MR02	DLP-RFID2	DLP-RFID1-0G			IPT-PP	IOT-F116-RAM-V1
Supply voltage (Vdc)	5 V	2.7-5.5 V	3.0-5.0 V		USB Port Powered from Host PC (USB 1.1/2.0 Compatible)	5V +/- 5%	20-30 V	17.5-30.6 V
Frequency	134.2 kHz	134.2 kHz	13.56 MHz		13.56 MHz	13.56 MHz	125 kHz	13.56 MHz
Communication Interface	I/O Logic	RS232(TTL), USB	TTL Serial Interface at 115,200 Baud	USB 1.1/2.0 Compatible, Mini-B 5-Pin Connector	1x RS232-TTL 1x Data/Clock-interface	RS-232, RS-485	Interface type depends on base used	RS 485
Standard	ISO 11785	ISO 11784/11785/14223 (HDX)	ISO 15693, 18000-3, 14443A/B and FelCa	ISO15693, ISO18000-3, ISO14443-A, ISO14443-B, ISO15693		ISO18000-6/A/B, EPC UHF Class 1, EPC Gen 2	Protection rating IP67	Protection rating IP67
Required antenna inductance (µH)	115 to 117.0; 200 min	47		Built-in Antenna: Up to 4-Inch Read Range	Integrated - 48 x 48 mm		Integrated	Integrated
Transmission Principle	HDX/FSK	HDX/FSK						
Dimensions (mm)	60.2 x 55.1 x 11.5	27.9 x 22.8 x 3.5	41.91 x 18.669 x 4.318	5.1x55.1x79.3	50x50x 5	170x320x48	80x80x60	99x67x22
Package	Module	SMD	Module	Module	Module	Module	PBT housing	PBT housing
Operating Temperature Range (°C)	0 to 50	-40 to + 85	0 to 70	0 to 70	-20 up to 70	-25 up to 55	-25 up to 70	-10 up to 55

6. JTC 1/SC 31: Automatic identification and data collection techniques

RFID limitations

Cost. The cost has gone down in recent years but it still needs to decrease further to meet the requirement of both low-end and high-end solutions. Prices of active or semi-passive tags limit their application to scanning of high-value goods over long ranges.

Standardisation. Standards for various groups of applications have evolved but they are a little sparse and leave much freedom in the choice of communication protocols and the format and amount of information stored in the tag. This can cause disagreement between various companies providing solutions based on RFID.

Collision. Anti-collision algorithms that can avoid signal collision while reading several tags at a time are either patented or patent pending, which adds to the cost of RFID-based system development.

Obstacle for RFID signals. Properties of some conductive materials such as goods containing water, or metal surfaces may be an obstacle to RFID application at a given frequency, as they may corrupt data transmission either by absorption or by ambient reflection of the signals.

Manufacturing not 100 per cent perfect. Manufacturing of tags is not 100 per cent failure-free even today. There is still a high percentage of defective tags in each lot.

Quick technology obsolescence. One of the major concerns of companies making RFID solutions is quick obsolescence of technology. With rapidly growing technology, more fault-tolerant readers outdate the older ones very quickly.

Security and privacy issues. Security and privacy is still a discussion for RFID technology. Proper encryption must be ensured at all interfaces where data could be intercepted or transmitted.

Possible attacks. Though there are no such attacks reported, the current RFID software can be vulnerable as reflected by various studies. ●

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